

Pregnancy-Related Nausea and Vomiting (N/V) and Contraception

Based on Pharmacotherapy Handbook, 7th Edition

Prepared for Pharmacy Yr IV UG Students

Overview

- Pregnancy-related nausea and vomiting (NVP), commonly known as morning sickness, affects up to 80% of pregnancies, while effective contraception is essential for reproductive planning and maternal health.

NVP: Definition

- NVP is a self-limiting condition typically occurring in the first trimester.
- Severe, persistent cases are classified as **hyperemesis gravidarum**, which can lead to dehydration, weight loss, and electrolyte imbalance.

NVP: Pathophysiology

- Exact mechanism unknown.

- Linked to elevated human chorionic gonadotropin (hCG) and estrogen levels.

- Psychological and genetic predispositions may play a role.

- Gastric dysrhythmia and delayed emptying may contribute.

NVP: Clinical Presentation

- Nausea with or without vomiting, usually between 4–9 weeks gestation.
- Symptoms peak at 9–16 weeks and usually resolve by 20 weeks.
- Severe form: hyperemesis gravidarum → dehydration, ketonuria, weight loss >5%.

NVP:
Nonpharmacologic
Management

- Dietary modification: small, frequent meals; avoid fatty/spicy foods.
- Avoid triggers (odors, heat, motion).
- Adequate hydration and rest.
- Ginger (250 mg QID) and acupressure may reduce symptoms.

NVP: Pharmacologic Therapy

Stepwise approach:

1. First-line: Pyridoxine (Vitamin B6) 10–25 mg q8h ± Doxylamine 12.5 mg.

2. Second-line: Antihistamines (diphenhydramine, meclizine), dopamine antagonists (metoclopramide, promethazine).

3. Third-line: Ondansetron or corticosteroids for refractory cases.

→ Avoid thalidomide and other teratogenic agents.

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Algorithm:
stepwise NVP
management.

- **Step 1: Initial Assessment**
 - Rule out other causes (e.g., gastroenteritis, UTI, thyroid dysfunction).
 - Assess severity using tools like PUQE score.
 - Evaluate hydration, weight loss, and ketonuria.
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Algorithm:
stepwise NVP
management...

- **Step 2: Lifestyle and Dietary Modifications**
- Eat small, frequent meals.
- Avoid triggers (strong smells, spicy/fatty foods).
- Use ginger (250 mg 3–4× daily).
- Acupressure (e.g., P6 wrist bands).

Algorithm:
stepwise NVP
management...



Step 3: First-Line Pharmacologic Therapy



Vitamin B6 (Pyridoxine):
10–25 mg orally 3–4×
daily.



± Doxylamine: 10–25 mg
orally 3–4× daily.



Can be combined or used
as delayed-release
formulation.

Step 4: Second-Line Options

If symptoms persist:

Antihistamines:

- Dimenhydrinate 25–50 mg q4–6h PO or PR.
- Meclizine or Cyclizine 25 mg PO.

Dopamine antagonists:

- Metoclopramide 5–10 mg PO/IV q8h.
- Prochlorperazine or Promethazine.

Algorithm:
stepwise NVP
management...

Algorithm: stepwise NVP management...

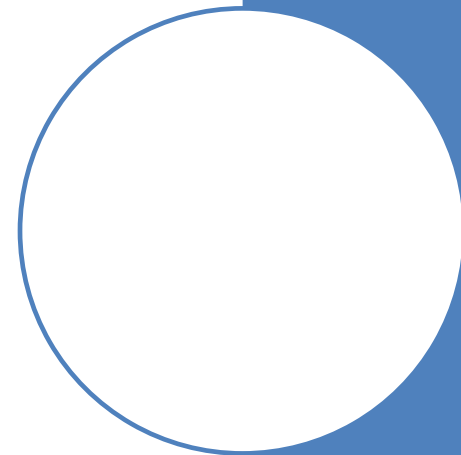
Step 5: Third-Line Therapy

Serotonin antagonists:

- Ondansetron 4–8 mg PO/ODT q8h (use cautiously due to safety concerns).

Corticosteroids:

- Methylprednisolone 16 mg PO q8h × 3 days (only in refractory cases and after 10 weeks gestation).



Algorithm:
stepwise NVP
management...

- **Step 6: Supportive Measures**
 - **IV fluids** (NS or LR) for dehydration.
 - **Thiamine supplementation** (especially before glucose infusion).
 - **Electrolyte correction.**
 - **Enteral or parenteral nutrition** in severe hyperemesis gravidarum.

Algorithm:
stepwise NVP
management...

- **Step 7: Monitoring and Follow-Up**
 - Reassess PUQE score and weight.
 - Monitor for complications (Wernicke's encephalopathy, electrolyte imbalance).
 - Consider hospitalization if outpatient management fails.

Hyperemesis Gravidarum: Management

- IV fluid and electrolyte correction.

- Thiamine supplementation before glucose infusion.

- Parenteral or enteral nutrition if oral intake inadequate.

- Antiemetics as per severity.

Contraception: Overview

Contraceptive methods prevent unintended pregnancy through

- suppression of ovulation,
- barrier methods, or
- implantation inhibition.

Choice depends on individual medical history, reproductive goals, and contraindications.

Hormonal Contraceptives

1. Combined estrogen-progestin methods:

- Oral pills, transdermal patches, vaginal rings.

- Mechanism: inhibit ovulation, thicken cervical mucus.

2. Progestin-only methods:

- Pills (norethindrone), injections (depot medroxyprogesterone), implants (etonogestrel), IUD (levonorgestrel).

- Safe for lactating women.

Nonhormonal Methods

- Copper IUD – long-term, hormone-free, highly effective.

- Barrier methods – condoms, diaphragms, cervical caps.

- Spermicides – nonoxynol-9, used alone or with barriers.

- Fertility awareness methods – cycle tracking and abstinence during fertile period.

Emergency Contraception

Used after unprotected intercourse or contraceptive failure.

- Levonorgestrel (Plan B): 1.5 mg single dose within 72 h.

- Ulipristal acetate: 30 mg within 120 h.

- Copper IUD: Most effective if inserted within 5 days.

→ Not for routine use; provide counseling on regular contraception.

HORMONAL REGULATION OF OVULATION

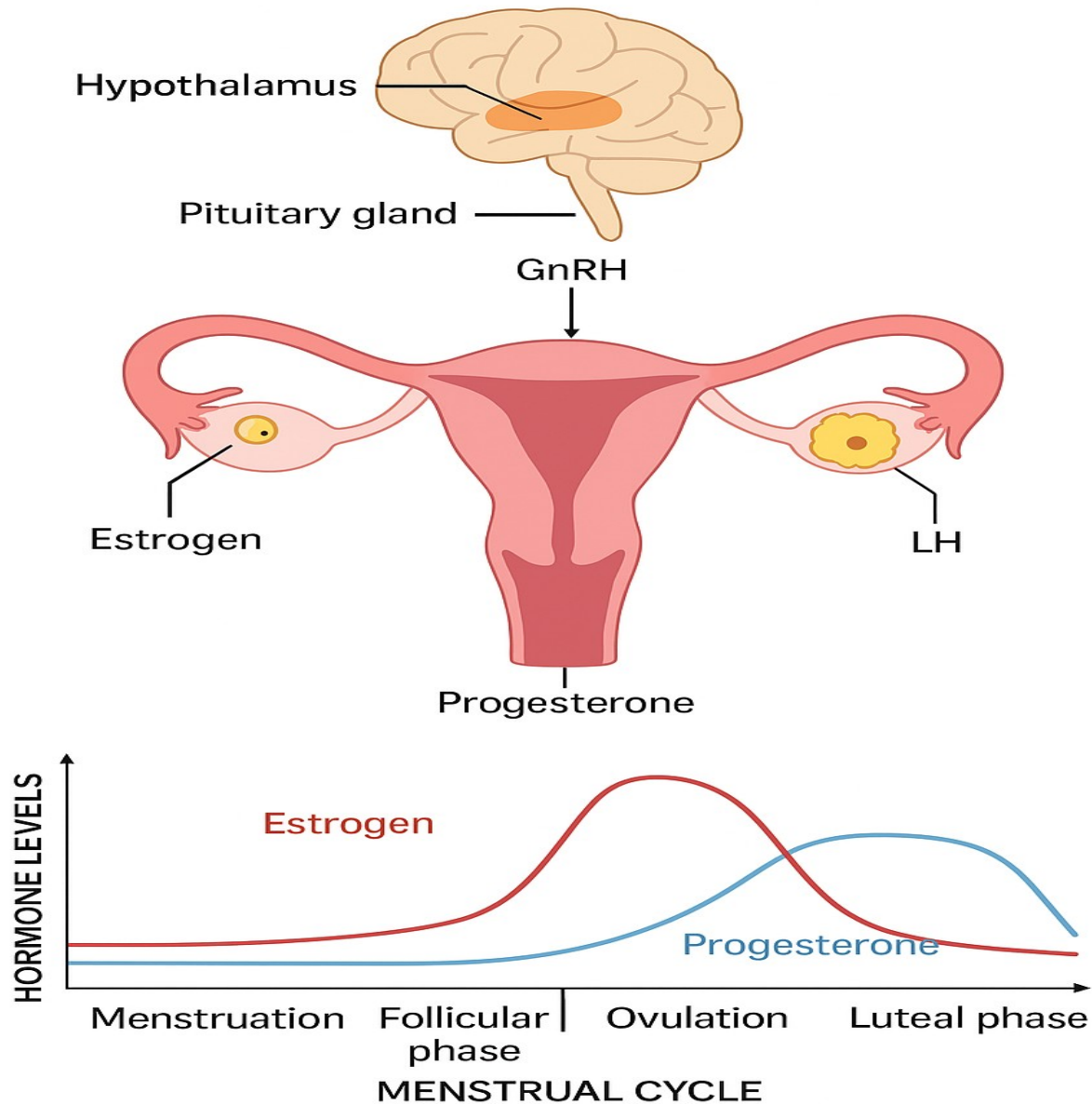
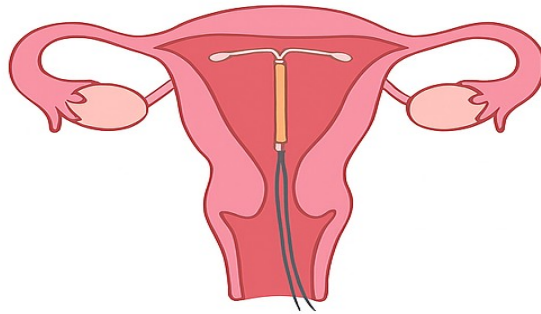
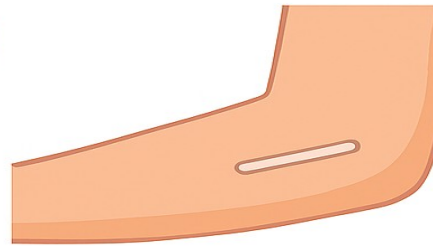


Chart:
contraceptive
efficacy
comparison.

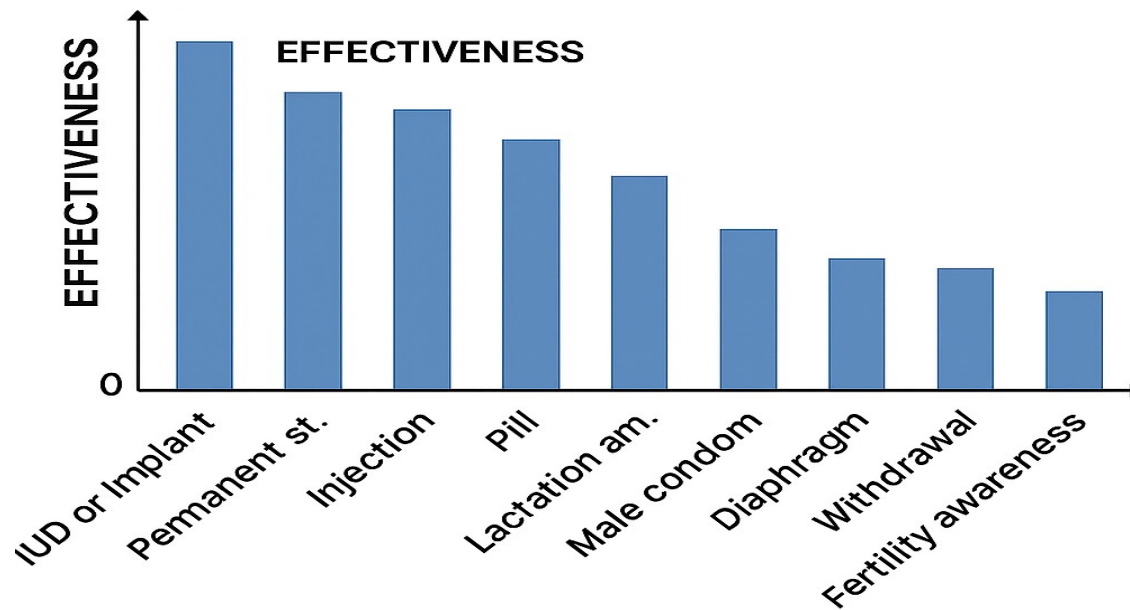
Method	Typical Use Failure Rate (%)	Perfect Use Failure Rate (%)
Male Condom	13	2
Female Condom	21	5
Oral Contraceptive Pills	7	0.3
IUD (Copper)	0.8	0.6
IUD (Hormonal)	0.1	0.1
Contraceptive Implant	0.1	0.05
Contraceptive Patch	7	0.3
Injectable Contraceptives	4	0.2
Fertility Awareness Methods	24	0.4
Emergency Contraception	20	5



**Copper or
Hormonal IUD**



Hormonal implant



Summary



Nausea & vomiting common in early pregnancy (4–12 weeks).



Etiology: ↑ hCG, estrogen, progesterone-induced delayed gastric emptying.



Non-pharmacologic management: small frequent meals, ginger, vitamin B6.



Pharmacologic: B6 + doxylamine (first-line), antihistamines, metoclopramide, ondansetron (severe).



Hyperemesis gravidarum: dehydration, electrolyte imbalance → requires IV fluids.

Summary...

- Contraception in Pregnancy Context:

Pre-conception: folic acid + effective contraception until planned pregnancy.

Safe postpartum options: POPs, injectables, implants, IUDs.

Avoid combined OCPs immediately postpartum in breastfeeding mothers.

References

Pharmacotherapy Handbook,
7th Edition – Chapters on
Nausea/Vomiting in Pregnancy
and Contraception.

ACOG Guidelines on NVP and
Contraceptive Use.

Recent WHO and CDC
publications on reproductive
pharmacotherapy.